

## Article

# Reliability Evaluation of Integrated Electricity–Water System Based on Multi-State Models of Equipment in Water System

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**Abstract:** The power system and the water system are two important infrastructures of human society that are closely related and interdependent. However, the reliability problems of the power system and water system are becoming more and more prominent. To better reveal the impact of the complex coupling relationship between the power system and the water system on the reliability of the Integrated Electricity–Water System (IEWS), this paper investigates a reliability evaluation method of the IEWS based on multi-state models of equipment in the water system. Firstly, a multi-state reliability model is established based on the failure mechanism of equipment in the water system, such as pipes and Reverse Osmosis (RO) desalination plants. Secondly, combined with the multi-state model of equipment in the water system and the Markov chain Monte Carlo (MCMC) method, the IEWS reliability evaluation method is established. Finally, two IEWSs with different scales are simulated to verify the validity and adaptability of the proposed model.

**Keywords:** integrated electricity–water system; Markov chain Monte Carlo; reliability evaluation; RO desalination plant; water system



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## 1. Introduction

The power system and the water system (WS) are closely related and deeply coupled. The Integrated Electricity–Water System (IEWS) has attracted great attention recently due to its merits of high efficiency, cost-effectiveness, and low carbonization [1–3].

However, the reliability problems of the power system and the WS are becoming more and more prominent, and the IEWS's deep connection highlights its reliability problems. During the peak of the 2021 Texas blizzard, up to 4.5 million households lost electricity, and the shortage of electricity had a cascading impact on the water supply system, with around 13 million people facing water shortages [4]. The electricity consumption of the water system accounts for 10~20% [5–8] of the total electricity consumption, which is an important electricity load. However, leaks in water supply pipes not only waste water resources, but also put extra operating pressure on the power system, which in turn affects power system reliability to some degree. For example, the WS experiences significant leakage issues in Brazil and California, which cause the pumps to consume a higher amount of power, consuming 1–4% more electricity than under normal operation conditions [9]. Furthermore, despite the high energy consumption of desalination technology of about 3.5–4.5 kWh/m<sup>3</sup>, the application of desalination technology is inevitable in certain regions where seawater is the only available water source, such as the Middle East and North Africa. In these regions, the demand for desalination has increased exponentially along

with the demand for water, which is used not only for drinking, but also for irrigation and industry, for example, providing cooling water for power plants. Therefore, the failure of equipment in the water system may lead to a significant increase in the overall power consumption of the water system, thereby increasing the operating pressure of the power system, and it may also lead to an insufficient supply of cooling water for thermal power generating units, with component failure spreading to the power system, thereby affecting the reliability of the power system [10].

In this situation, it is imperative to propose a reliability evaluation approach for the IEWS in order to guarantee the safe operation of the system and to give system planning and operation a quantitative foundation for decision making. The failure mechanisms of equipment in the WS are very complex, and there is strong coupling between the power system and water system; therefore, the key challenge to address is how to construct an accurate equipment reliability model for the WS to accurately quantify the reliability of the electricity and water supply. However, research on IEWS reliability evaluation is lacking.

The WS involves water production, transmission, and storage, as well as key equipment such as desalination plants, water pipes, pumps, reservoirs, and water valves [11]. Reference [11] investigated the mechanical and hydraulic reliability of the WS by using the minimum cut-set method, which establishes a normal-shutdown model of water supply pipes. However, the water supply capacity of the WS is closely related to the leakage state of water supply pipes [7], and the two-state model obviously cannot accurately portray the actual operating conditions of the system, overestimating WS reliability. References [12,13] analyzed the optimal scheduling of the IEWS, but the model of RO desalination is simple and did not establish a multi-state reliability model for RO desalination plants, ignoring the impact of component failure in RO desalination on IEWS optimal scheduling. Reference [14] considered the coupling characteristics of the IEWS and analyzed the vulnerability of the power system, but it did not consider the reliability indices of the IEWS, making it difficult to ascertain the level of reliable energy supply of the IEWS. By conducting simulation experiments on a realistic IEWS, the authors of [15] have analyzed the impact of cascading failures of the power system on water supply shortages under different maximum power line capacities and water demand conditions, and they have demonstrated that the urban WS has a strong dependence on the reliability of the power network. However, the article neglected the impact of WS failures on the reliability of the power system. It can be seen that there is a lack of effective IEWS reliability evaluation methods to quantify the level of IEWS electric and water load supply reliability.

To solve the above problems, this paper investigates the IEWS reliability evaluation approach, considering multi-state models of equipment in the WS. The main contributions of this paper are as follows:

- (1) Establishing a multi-state reliability model for equipment in the WS based on their failure mechanisms to accurately assess the reliability level of the IEWS. This study investigates the failure mechanism of water supply pipes and creates a multi-state reliability model considering leaks. Furthermore, a multi-state reliability model of an RO desalination plant is established based on the process technology concept of the plant, taking into account inefficient operation.
- (2) Modeling the optimal load shedding of the IEWS and establishing reliability indices. An optimal load shedding model of the IEWS considering water–energy coupling is proposed, and a reliability evaluation index system for the IEWS is established in terms of the probability of the loss of electric and water loads when the expected value of electricity and water is not supplied.
- (3) Modeling IEWS reliability models based on Markov chain Monte Carlo (MCMC) simulation. An IEWS reliability evaluation approach based on MCMC simulation is

established to effectively quantify the reliability level of electric and water supply in the IEWS. This approach serves as a basis for rational and scientific decision making during the system's design and construction, as well as for successfully quantifying the reliability level of the IEWS.

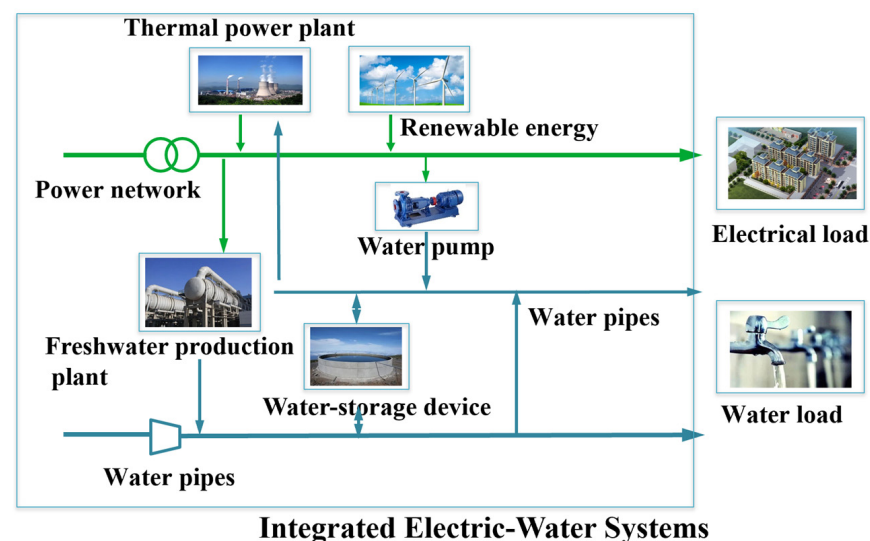
The rest of the paper is organized as follows. Section 2 proposes the multi-state reliability models of equipment in the WS, Section 3 proposes an optimal load shedding model of the IEWS and a reliability evaluation index system, and Section 4 discusses the proposed IEWS reliability evaluation model and solution method based on the multi-state models of equipment in the WS. Section 5 presents the numerical case study and Section 6 concludes with the research focus, research methods, and conclusions of this paper.

## 2. Reliability Model of the Components in the WS

In this section, the structure of the IEWS is introduced, and the operational characteristics and failure mechanisms of equipment in the WS are analyzed. Then, multi-state reliability models for water pipes and the RO desalination plant considering leakage and inefficient operation are proposed.

### 2.1. Structure of the IEWS

The structure of the IEWS is shown in Figure 1, where the desalination plant turns salt water into fresh water by consuming electrical energy. In addition, the water pumps also consume electrical energy to keep the WS operating properly. Therefore, the WS is tightly coupled with the power system through the desalination plant and the pumps.



**Figure 1.** Typical structure diagram of the IEWS.

The regulation of the desalination process can improve the operational flexibility of the IEWS. In the peak period of renewable energy output or the valley period of power demand, the desalination plant's fresh water production can be increased, and the water will be transported to the reservoir for storage. On the contrary, the fresh water generation of the desalination plant can be reduced as well as the pumping power of the water pumps. The stored water in the reservoirs of the desalination plant and tanks of the distribution pipeline network can be released to maintain the balance between supply and demand in the WS.

## 2.2. Multi-State Markov Modeling of Desalination Plants

The RO desalination method is the simplest, most economical, and most efficient of the existing desalination technologies, and its popularity has been increasing in recent years [16,17]. In this paper, the RO desalination method is taken as an example to model the desalination plant.

The RO desalination system uses the selective permeability of semi-permeable membranes to realize seawater desalination. The basic process flow is shown in Figure 2. The feed seawater is firstly sent into the pre-treatment unit by the intake pump, and it then enters into the clean water tank after flocculation, sedimentation, filtration, etc. The treated seawater is then pressurized by the high-pressure pump of the RO desalination device to force the water molecules to enter into the fresh water side through semi-permeable membranes to realize the separation of salt water. Finally, the product water is formed after water quality adjustment and other after-treatment processes.

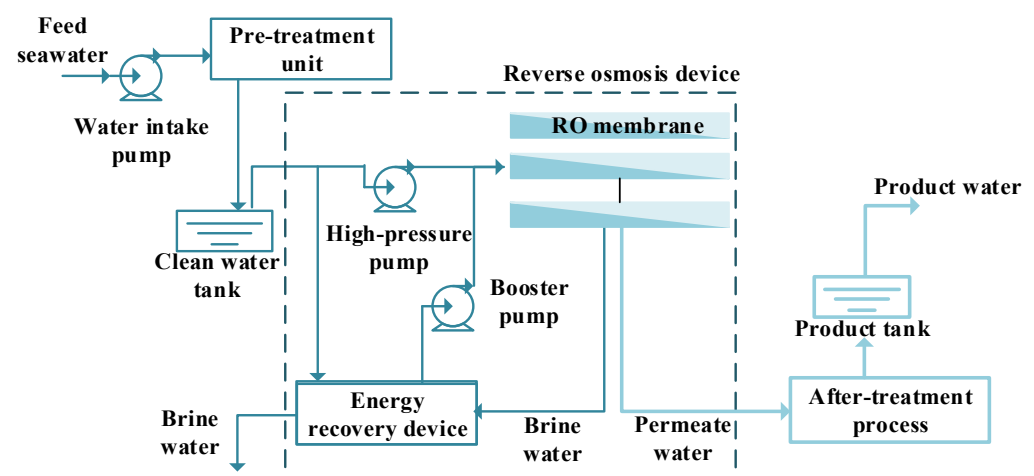


Figure 2. Process flow diagram of RO desalination plant.

The water intake pump is one of the major energy consuming devices in RO desalination plants [2]. Assuming that the hydrostatic head of the water intake pump is fixed, the operating constraint is as follows:

$$\begin{cases} P_t^{\text{in}} = \frac{\rho g}{\eta^{\text{in}}} q_t^{\text{in}} H_t^{\text{in}} \\ q_t^{\text{in}, \min} \phi_t^{\text{in}} \leq q_t^{\text{in}} \leq q_t^{\text{in}, \max} \phi_t^{\text{in}} \end{cases} \quad (1)$$

where  $P_t^{\text{in}}$ ,  $q_t^{\text{in}}$ , and  $H_t^{\text{in}}$  represent the power consumption, water flow, and hydrostatic head of the water intake pump at time  $t$ , respectively;  $\rho$  is the density of seawater;  $g$  is the acceleration of gravity; and  $\eta^{\text{in}}$  is the efficiency of the pump. In this paper,  $\phi$  always represents the working state of the equipment, which is a 0–1 variable (take 1 to indicate the operation), and will not be repeated below.

The RO device is composed of high-pressure pumps, booster pumps, energy recovery devices, and several RO membranes. The inlet and outlet volumes of the RO units and the power they consume satisfy the following (2):

$$\begin{cases} q_{r,t}^{\text{out}} = K_r A_r (p_{r,t} - \Delta\pi) \\ q_{r,t}^{\text{in}} = q_{r,t}^{\text{out}} / R \\ P_{r,t}^{\text{RO}} = q_{r,t}^{\text{out}} [a_r c_r^{\text{in}} \frac{2-R}{2(1-R)} + b_r (p_{r,t} - \Delta\pi)] \\ \Delta p_r^{\min} \phi_{r,t} \leq p_{r,t} - \Delta\pi \leq \Delta p_r^{\max} \phi_{r,t} \end{cases} \quad (2)$$

where  $q_{r,t}^{\text{in}}$  and  $q_{r,t}^{\text{out}}$  are the water inlet and outlet of the RO unit;  $P_{r,t}^{\text{RO}}$  is the power consumed by the RO unit;  $p_{r,t}$  is the working pressure of the RO unit;  $K_r$ ,  $A_r$ , and  $\Delta\pi$  are the permeability coefficients of the osmosis membrane, the permeable area, and the difference in permeability pressure across the surface area of the membrane, respectively; and  $R$  is the recovery rate of the RO.

Assuming that the number of RO membranes of the RO unit is  $r$ , the total water inlet and outlet of the RO unit, the power consumed by the high-pressure pump, as well as the booster pump, satisfy the following relationship:

$$\begin{cases} q_{\text{RO},t}^{\text{out}} = \sum_{r \in \text{RO}} q_{r,t}^{\text{out}} \\ q_{\text{RO},t}^{\text{in}} = \sum_{r \in \text{RO}} q_{r,t}^{\text{in}} \\ P_{\text{RO},t}^{\text{RO}} = \sum_{r \in \text{RO}} P_{r,t}^{\text{RO}} \\ q_{\text{RO},t}^{\text{in}} \leq q_t^{\text{in}} \beta \end{cases} \quad (3)$$

where  $q_{\text{RO},t}^{\text{in}}$ ,  $q_{\text{RO},t}^{\text{out}}$  and  $P_{\text{RO},t}^{\text{RO}}$  represent the total inlet water volume of the RO unit, the total outlet water volume, and the total power consumed by the high-pressure pump and the booster pump, respectively;  $\beta$  refers to the ratio of the amount of seawater after pre-treatment to the amount of feed water.

The clear and product tank are water storage facilities with the following operational constraints:

$$\begin{cases} h_t^{\text{CWP}} = h_{t-1}^{\text{CWP}} + (q_t^{\text{in}} \beta - q_{\text{RO},t}^{\text{in}}) \Delta t / A^{\text{CWP}} \\ h_t^{\text{PRP}} = h_{t-1}^{\text{PRP}} + (q_{\text{RO},t}^{\text{out}} - q_t^{\text{out}}) \Delta t / A^{\text{PRP}} \\ h^{\text{CWP},\text{min}} \leq h_t^{\text{CWP}} \leq h^{\text{CWP},\text{max}} \\ h^{\text{PRP},\text{min}} \leq h_t^{\text{PRP}} \leq h^{\text{PRP},\text{max}} \end{cases} \quad (4)$$

where  $h_t^{\text{CWP}}$  and  $h_t^{\text{PRP}}$  are the water storage capacity of the clear tank and product tank;  $A^{\text{CWP}}$  and  $A^{\text{PRP}}$  are the bottom area of the clear tank and product tank;  $h^{\text{CWP},\text{max}}$  and  $h^{\text{PRP},\text{max}}$  are the maximum capacity of the clear tank and product tank; and  $h^{\text{CWP},\text{min}}$  and  $h^{\text{PRP},\text{min}}$  are the minimum capacity of the clear tank and product tank.

The multi-state reliability model can be established by analyzing the failure mechanism of the desalination plant. The details are as follows:

The desalination plant operates regularly when all its components work as usual, as shown in Figure 2. The consumed electricity power and the produced desalinated water are calculated using Equations (1)–(4). The desalination plant cannot operate normally if the water intake pump breaks. If the clear water tank fails, the RO desalination plant cannot desalinate without feed water. Suppose the high-pressure pump fails and cannot pressurize the water; the permeability pressure forces the permeated water to flow back to the feed water side through the membrane, and the RO unit cannot work correctly. If the product tank and after-treatment process fail, the desalination plant cannot produce water. Therefore, the RO desalination plant is in a fault shutdown state when any of the above components (referred to as component 1) fail.

Suppose the pre-treatment unit fails; it will not be able to perform flocculation, precipitation, filtration, and the other processes of the treatment of seawater. The concentration of seawater in the RO unit will be higher;  $\Delta\pi$  will be bigger;  $(p_{r,t} - \Delta\pi)$  will be smaller, which can be derived from (2) and which represents the desalination water flow rate  $q_{r,t}^{\text{out}}$  through the RO unit is smaller. So, the RO desalination plant is in inefficient operation. In addition, the failure of the pre-treatment unit leads to an increase in the concentration  $c_r^{\text{in}}$  of the feed seawater, which results in an increase in the power  $P_{r,t}^{\text{RO}}$  consumed by the

high-pressure pump. Therefore, in the case of the failure of any of the components mentioned above (denoted as component 2), the RO desalination plant will enter an inefficient operation state.

Suppose the booster pump and energy recovery device fail; a reduction in RO recovery  $R$  will result in a reduction in desalinated water flow  $q_{r,t}^{\text{out}}$  through the RO unit, and the desalination plant will be in a state of inefficient operation. This component is identified as component 3.

In the RO unit, over time, the RO membrane will gradually age, fouling and other phenomena will occur, and the permeation efficiency will be reduced, leading to a decrease in the permeation coefficient  $K_r$  of the RO membrane. The desalination plant will be in an inefficient operation state. RO membrane aging and the fouling phenomenon are referred to as component 4. Over time, the RO membrane will fail due to the immense pressure difference between the two sides, decreasing the number of effective RO membranes  $r$ . The high-pressure pumps must increase electricity consumption to maintain the RO unit's water supply and ensure fresh water production. The desalination plant will be in an inefficient operation state. Therefore, the failure of the RO membrane is labeled as component 5, which will result in the inefficient operation of the RO unit.

The multi-state Markov model of the RO desalination plant is established by analyzing the above failure mechanism, as shown in Figure 3. In the model,  $\lambda_{\text{RO},1-5}$  and  $\mu_{\text{RO},1-5}$  are the transition rates of the RO desalination plant; “1U, 2U, 3U, 4D, 5D” indicate that components 1, 2, and 3 are in the normal working state, and components 4 and 5 are in the fault state, respectively.

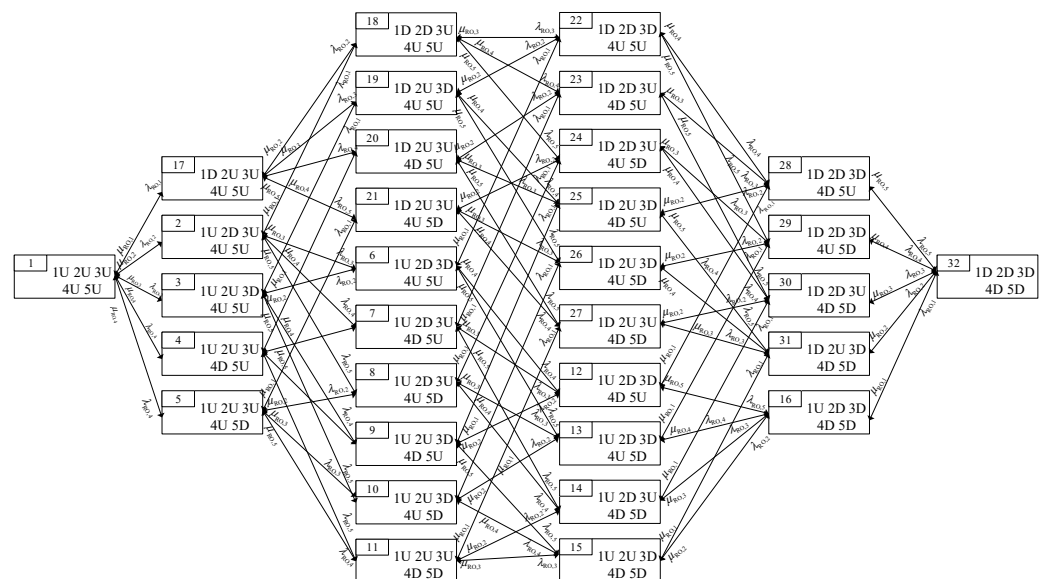


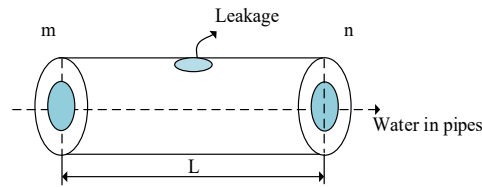
Figure 3. Multi-state Markov model of RO seawater desalination plant.

The operating state of the RO desalination plant can be divided according to Figure 3, where the operating state variables of the RO desalination plant are  $s_{\text{RO},t}$ ,  $s_{\text{RO},t} = 1, 2, \dots, 32$ ;  $P_{\text{RO}} = (1 + RP_{s_{\text{RO},t}})g(q_{s_{\text{RO},t}}^{\text{out}})$  denotes the functional relationship between the power consumption  $P_{\text{RO}}$  and the water production  $q_{s_{\text{RO},t}}^{\text{out}}$  of the desalination plant, which can be obtained according to (1) to (4); and  $RP_{s_{\text{RO},t}}$  is the proportion of the increase in power consumption when the desalination plant is in the state  $s_{\text{RO},t}$  [2].

### 2.3. Multi-State Markov Modeling of Water Supply Pipes

The reliable operation of the WS is essential to ensure an uninterrupted supply of clean water. Pipeline leakage is a significant factor affecting the proper operation of the WS.

Studies have shown that pipeline leakage is caused by a complex interaction of physical, environmental, and operational factors that affect the rate of deterioration, degree of displacement deformation, and rupture, and water supply pipes fail in various forms, including pinholes, holes, and ruptures, all of which can lead to water loss [18,19]. A schematic diagram of water supply pipe leakage is shown in Figure 4.



**Figure 4.** Structural diagram of a leaking water supply pipe.

Typically, the nodal balance of the WS under normal operating conditions is shown as follows (5):

$$\mathbf{A}\mathbf{Q}_P = \mathbf{Q}_N \quad (5)$$

where  $\mathbf{Q}_P = [q_{Pj}]$  ( $\text{m}^3/\text{s}$ ) is a vector, each element of which represents the amount of water transported in the pipe;  $\mathbf{Q}_N = [q_{Nj}]$  ( $\text{m}^3/\text{s}$ ) is a  $n \times 1$  vector, each element of which represents the node water demand; and  $A$  is the connection matrix of pipes and nodes.

When a fault in a water supply pipe results in a leak, assuming that after the leakage occurs it takes only a relatively short period to reach a steady-state leakage, (5) will be modified to (6), as follows:

$$\mathbf{A}\mathbf{Q}_P = \mathbf{Q}_N + \mathbf{Q}_L \quad (6)$$

where  $\mathbf{Q}_L = [q_{Lj}]$  ( $\text{m}^3/\text{s}$ ) is a  $n \times 1$  vector and each element represents the leakage flow rate of the node. The following point leakage model is proposed based on the theory of leakage flow in the WS as follows [20]:

$$q_{Lj} = \alpha \mu A_{Lj} \sqrt{2gH_{Lj}} \quad (7)$$

where  $H_{Lj}$  (m) is the leakage point hydrostatic head;  $\mu$  is the flow coefficient;  $\alpha$  is the leakage coefficient related to the obstruction of the soil and filler, usually 0.1–0.3;  $q_{Lj}$  is the leakage flow; and  $A_{Lj}$  ( $\text{m}^2$ ) is the leakage area, which is calculated as follows.

$$A_{Lj} = \begin{cases} 0, S < R \\ \pi D(S - R), R \leq S \leq R + \frac{A}{\pi D} \\ A, S > R + \frac{A}{\pi D} \end{cases} \quad (8)$$

where  $R$  (m) is the joint leakage displacement, i.e., the joint begins to leak when the joint displacement arrives at  $R$ . Here, displacement refers to the size of the damage point of the pipeline.  $D$  (m) is the diameter of the pipeline.  $A$  ( $\text{m}^2$ ) is the cross-sectional area of the pipeline,  $S$  (m) is the displacement of the pipeline after its failure, which is mainly related to the degree of deformation or rupture of the pipeline.

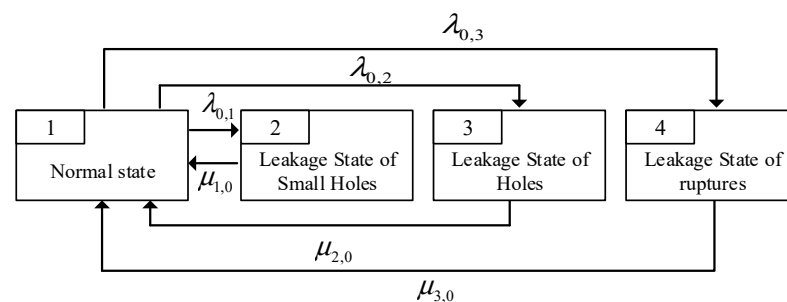
The leakage volume calculation can use the virtual node method [18], but the method is complex and difficult to calculate. In order to simplify the calculation, reference [19] assumes that the leakage of the pipe comes from its two endpoints, which significantly improves the efficiency of the calculation, and the leakage results are close to other methods. The leakage flow rate of the pipe is as follows:

$$q_{Lj} = \alpha \mu A_{Lj} \sqrt{g(H_1 + H_2)} \quad (9)$$

where  $H_1$  and  $H_2$  are the hydrostatic heads at the two end points of the leaking pipe, and the hydrostatic heads are calculated according to Equation (17) using the pressure of the pipe nodes. Due to the randomness of pipeline leakage, the Monte Carlo sampling method is used to extract the leakage aperture of the pipeline, the leakage area is calculated according to Equation (8), and the leakage flow rate is calculated according to Equation (9).

In existing studies, two-state reliability evaluation models are generally used for water supply pipes containing normal operating states and fault states [11,21]. However, the size of leak displacements is random, including normal, small hole leakage, hole leakage, and rupture. The normal and other operation states can be transformed into each other. In contrast, the fault state can only be transformed into the normal operation state since the pipeline will recover to normal operation after detecting leakage.

In this paper, it is assumed that only one state change occurs in the sampling time interval  $\Delta t$  (1 h). A multi-state Markov model of water supply pipes is established according to the failure mechanism analysis of water supply pipes, as shown in Figure 5. In the figure,  $\lambda_{0,1}$ ,  $\lambda_{0,2}$ , and  $\lambda_{0,3}$  represent the transition rates between the normal state and the state of small hole leakage, hole leakage, and rupture, respectively, and  $\mu_{1,0}$ ,  $\mu_{2,0}$ , and  $\mu_{3,0}$  represent the repair rate of the water supply pipe.

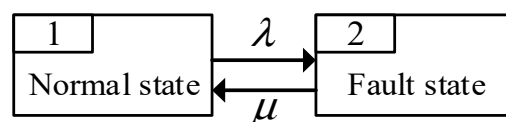


**Figure 5.** Multi-state Markov model of water supply pipes.

According to Figure 5, the operation state of the water supply pipelines can be divided. Define the operating state variable of water supply pipeline as  $s_{L,t}$ ,  $s_{L,t} = 1, 2, 3, 4$ ;  $q_{Lj,t} = f_{s_{L,t}}(S_{s_{L,t}})$  denotes the functional relationship between the leakage amount  $q_{Lj}$  and the pipeline fault displacement  $S_{s_{L,t}}$  when the water supply pipeline is in the state  $s_{L,t}$ , which can be obtained according to (6) to (9).

#### 2.4. Two-State Markov Model for Water Storage Tanks and Valves

Water storage tanks are mainly used to store and provide water. Valves are mainly used for opening and closing pipelines and controlling the flow direction. This paper uses the classical two-state model, as shown in Figure 6. In the figure,  $\lambda$  and  $\mu$  represent the transfer rates between the two states of the water tank or valves.



**Figure 6.** A two-state Markov model of a water storage tank or valve.

### 3. Optimal Load Shedding Model for the IEWS

An optimal load shedding model of the IEWS, which considers the water–energy coupling, is established to simulate the system operation status and the shedding of the electric and water loads under the fault condition.

### 3.1. Objective Function

In order to guarantee the efficient consumption of renewable energy and the reliable supply of electric and water loads, the objective function of the IEWS's optimal load shedding model is to minimize the sum of load shedding penalty cost and the abandoned wind cost, which is shown in (10), as follows:

$$\min C = \sum_{i=1}^n \sum_{t=1}^T c_{en} \Delta D_{i,t}^{EDS} + \sum_{w=1}^{N_w} \sum_{t=1}^T c_{wc} P_{w,t}^{wc} + \sum_{Wm=1}^{N_{Wm}} \sum_{t=1}^T c_{Wm} \Delta D_{Wm,t}^{WDS} \quad (10)$$

where  $\Delta D_{i,t}^{EDS}$  and  $\Delta D_{Wm,t}^{WDS}$  are the curtailed loads of the power system and the WS; water load shedding  $\Delta D_{Wm,t}^{WDS}$  means that the water system is not providing the required amount of water to the water users;  $P_{w,t}^{wc}$  is the curtailed wind power; and  $c_{en}$ ,  $c_{wc}$ , and  $c_{Wm}$  are the unit electric and water load shedding penalty cost, and the unit abandoned wind penalty cost.

### 3.2. Power System Constraints

$$\sum_{i \in \Omega_n^T} P_{i,t} + \sum_{w \in \Omega_n^W} (P_{w,t}^{wn} - P_{w,t}^{wc}) - (D_{n,t}^{EDS} - \Delta D_{n,t}^{EDS}) - \sum_{h \in \Omega_n^{WDS}} P_{h,t}^{WDS} = \sum_{m \in \Lambda_n^E} B_{n,m} (\theta_{n,t} - \theta_{m,t}), \forall n \in \Lambda^E \quad (11)$$

$$|P_{i,t} - P_{i,t-1}| \leq R_i \phi_{i,t-1} + P_i^{gmin} (\phi_{i,t} - \phi_{i,t-1}) + P_i^{gmax} (1 - \phi_{i,t}), \forall i \in \Omega^T \quad (12)$$

$$P_i^{gmin} \phi_{i,t} \leq P_{i,t} \leq P_i^{gmax} \phi_{i,t}, \forall i \in \Omega^T \quad (13)$$

$$0 \leq P_{w,t}^{wc} \leq P_{w,t}^{wn}, \forall w \in \Omega^W \quad (14)$$

$$-S_{n,m} \leq B_{n,m} (\theta_{n,t} - \theta_{m,t}) \leq S_{n,m}, \forall (n, m) \in \Lambda^E \quad (15)$$

$$\theta_{ref,t} = 0 \quad (16)$$

where  $P_{i,t}$  is the node power of the power system,  $P_{w,t}^{wn}$  and  $D_{n,t}^{EDS}$  are the predicted value of the wind power and conventional electric load,  $P_{h,t}^{WDS}$  is the power consumed by the WS,  $n$  and  $m$  are both nodes of the water system,  $n$  represents the node where the water flows into the pipeline, and  $m$  represents the node where the water flows out of the pipeline.  $B$  represents the susceptance of transmission line,  $\theta$  represents the phase angle of the node, and  $R_i$  is ramp rate for thermal power generating unit.  $S_{n,m}$  is the transmission capacity.  $\Lambda_n^E$  is the collection of transmission system nodes,  $\Lambda^E$  is the collection of power transmission lines,  $\Omega^T$  is the collection of thermal power generating units, and  $\Omega^W$  is the collection of wind turbines.

Equation (11) is the nodal power balance constraint, (12) is the ramp rate constraint of the thermal power generating unit, (13) is the output power constraint of the thermal power generating unit, (14) is the wind abandonment power constraint, (15) is the power flow constraint of the power lines, and (16) is the phase angle constraint of the balancing node.

### 3.3. Water System Constraints

The WS constraints contain conventional constraints such as piping equations, pump equations, nodal flow balance equations, and upper and lower water flow limit constraints, in addition to the constraints of the desalination plant (1) to (9), which are modeled as follows:

$$H_{Wm,t} = \frac{P_{Wm,t}}{\rho g} + h_{Wm}, \forall Wm \in \Lambda^W \quad (17)$$

$$H_{W_{m,t}} - H_{W_{n,t}} = r_{W_n} \text{sgn}(q_{W_{mn,t}})(q_{W_{mn,t}})^2 \quad (18)$$

$$q_{W_{mn}}^{\min} \leq q_{W_{mn,t}} \leq q_{W_{mn}}^{\max}, \forall (W_m, W_n) \in \Delta^W \quad (19)$$

$$H_{P_{m,t}} = a_{P_m}(q_{P_{m,t}})^2 + b_{P_m}q_{P_{m,t}} + c_{P_m} \quad (20)$$

$$q_{P_m}^{\min} \phi_{P_{m,t}} \leq q_{P_{m,t}} \leq q_{P_m}^{\max} \phi_{P_{m,t}}, \forall P_m \in \Delta_P^W \quad (21)$$

$$(H_{W_{m,t+1}} - H_{W_{m,t}})A_{W_m}^{\text{Tank}} = W_{W_{m,t+1}} - (1 - \mu_W)W_{W_{m,t}} \quad (22)$$

$$W_{W_{m,t}} = (1 - \mu_W)W_{W_{m,t-1}} + (q_{W_{m,t}}^{\text{in}} - q_{W_{m,t}}^{\text{out}})\Delta t \quad (23)$$

$$W_{W_m}^{\min} \phi_{T_{m,t}} \leq W_{W_{m,t}} \leq W_{W_m}^{\max} \phi_{T_{m,t}}, \forall W_m \in \Lambda_T^W \quad (24)$$

$$-M(1 - \phi_{V_{mn,t}}) \leq H_{W_{m,t}} - H_{W_{n,t}} \leq M(1 + \phi_{V_{mn,t}}) \quad (25)$$

$$q_{W_{mn}}^{\min} \phi_{V_{mn,t}} \leq q_{W_{mn,t}} \leq q_{W_{mn}}^{\max} \phi_{V_{mn,t}}, \forall (W_m, W_n) \in \Delta_V^W \quad (26)$$

$$D_{W_{k,t}} - \Delta D_{W_{k,t}}^{\text{WDS}} = \sum_{W_m: (W_m, W_k) \in \Delta^W} q_{W_{mk,t}} - \sum_{W_k: (W_k, W_n) \in \Delta^W} q_{W_{kn,t}} - q_{LW_{mn,t}}, \forall W_k \in \Lambda^W \quad (27)$$

$$P_{P_{m,t}}^{\text{WP}} = \frac{\rho g}{\eta} q_{P_{m,t}} H_{W_{mn}}, \forall P_m \in \Delta_P^W, \forall (W_m, W_n) \in \Delta_P^W \quad (28)$$

$$P_{h,t}^{\text{WDS}} = P_t^{\text{in}} + P_{\text{RO},t}^{\text{RO}} + \sum_{P_m=1}^{N_{P_m}} P_{P_{m,t}}^{\text{WP}}, \forall h \in \Omega^{\text{WDS}} \quad (29)$$

where  $H_{W_{m,t}}$ ,  $P_{W_{m,t}}$ , and  $h_{W_m}$  represent the total hydrostatic head, stream pressure, and positional head at node  $W_m$  of the WS, respectively.  $\rho$  and  $g$  represent the density of water and gravitational acceleration, respectively.  $q_{W_{mn,t}}$  indicates the volume of water flowing through the pipe,  $r_{W_n}$  denotes the pipe head loss coefficient.  $H_{P_{m,t}}$  and  $q_{P_{m,t}}$  denote the head gain of pump  $P_m$  and the flow rate through the pump;  $a_{P_m}$ ,  $b_{P_m}$ , and  $c_{P_m}$  represent the head coefficient of the pump, respectively.  $q_{W_{m,t}}^{\text{in}}$ ,  $q_{W_{m,t}}^{\text{out}}$ , and  $W_{W_{m,t}}$  denote the inflow and outflow flow rates and the storage capacity of the cistern  $W_m$ , respectively;  $A_{W_m}^{\text{Tank}}$  and  $\mu_W$  denote the cross-sectional area of the reservoir and the rate of water loss in the impoundment, respectively.  $M$  is a very large constant.  $D_{W_{k,t}}$ ,  $q_{W_{mk,t}}$ , and  $q_{W_{kn,t}}$  are the water loads, inflow flows, and outflow flows at node  $W_k$  of the distribution network at the time;  $q_{LW_{mn,t}}$  is the leakage from the pipe if there is a fault in the pipe  $m - n$ .  $P_{P_{m,t}}^{\text{WP}}$  is the power consumed by the pump  $P_m$  and  $\eta$  is the pump efficiency.  $\Lambda^W$  represents the collection of water system nodes,  $\Delta^W$  represents the collection of water system pipes,  $\Delta_P^W$  is the collection of water pumps,  $\Lambda_T^W$  represents the collection of reservoirs, and  $\Delta_V^W$  is the collection of valves.

Equation (17) represents the total hydrostatic head constraint of the WS, (18) represents the along-track hydrostatic head loss of the pipeline (this paper used the Darcy-Weisbach equation [22]), (19) represents the pipeline flow constraint, (20) and (21) model the pumps [23,24], (22) to (24) are the reservoir constraints, (25) and (26) are the valve constraints, (27) is the node balance constraint, (28) is the pump power constraint, and (29) represents the WS power constraint.

It can be seen that the optimal load shedding model developed in this section is a typical nonlinear mixed integer programming problem. The nonlinear constraints (18), (20), and (28) are linearized using the segmented linearization method [23]; then, the model can be solved using Gurobi 9.5 commercial software.

## 4. Reliability Evaluation Method of Integrated Electric–Water System

### 4.1. Reliability Indices

The power system's reliability indices are already fairly developed, but there are currently no reliability indices that account for leakage and water supply sustainability. In

this section, the reliability indices of the WS are proposed by illustrating the expectation and probability of a load supply shortage.

(1) Loss of Electricity Load Probability (LOELP)

$$LOELP = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} x_{yt}^{EDS} \right) / 8760N_y \quad (30)$$

where  $x_{yt}^{EDS}$  is the state variable of the electric load shedding of the system at moment  $t$  in hours. If  $x$  is 1, it means that there is load shedding, and if  $x$  is 0, it means that there is no load shedding.

(2) Expected Energy Not Supplied (EENS)

$$EENS = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} \sum_{i=1}^n \Delta D_{i,yt}^{EDS} \right) / N_y \quad (31)$$

where  $\Delta D_{i,yt}^{EDS}$  is the electrical load shedding at node  $i$  of the power system at time  $t$  in hours.

(3) Loss of Water Load Probability (LOWLP)

$$LOWLP = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} x_{yt}^{WDS} \right) / 8760N_y \quad (32)$$

where  $x_{yt}^{WDS}$  is the state variable of water load shedding of the system at moment  $t$  in hours.

(4) Expected Water Not Supplied (EWNS)

$$EWNS = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} \sum_{m=1}^{W_m} \Delta D_{W_m,t}^{WDS} \right) / N_y \quad (33)$$

where  $\Delta D_{W_m,t}^{WDS}$  is the water load shedding at node  $W_m$  of the WS at time  $t$  in hours.

(5) Probability of Water Leakage (PWL)

$$PWL = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} x_{yt}^L \right) / 8760N_y \quad (34)$$

where  $x_{yt}^L$  is the state variable for the presence of pipe leakage in the WS at moment  $t$  in hours.

(6) Expected Water Leakage (EWL)

$$EWL = \left( \sum_{y=1}^{N_y} \sum_{t=1}^{8760} \sum_{i=1}^{N_{Wmn}} q_{LWmn,yt} \right) / N_y \quad (35)$$

where  $q_{LWmn,yt}$  is the leakage at pipe  $m - n$  of the WS at time  $t$  in hours.

(7) Comprehensive Loss of Energy Not Supply (CENS)

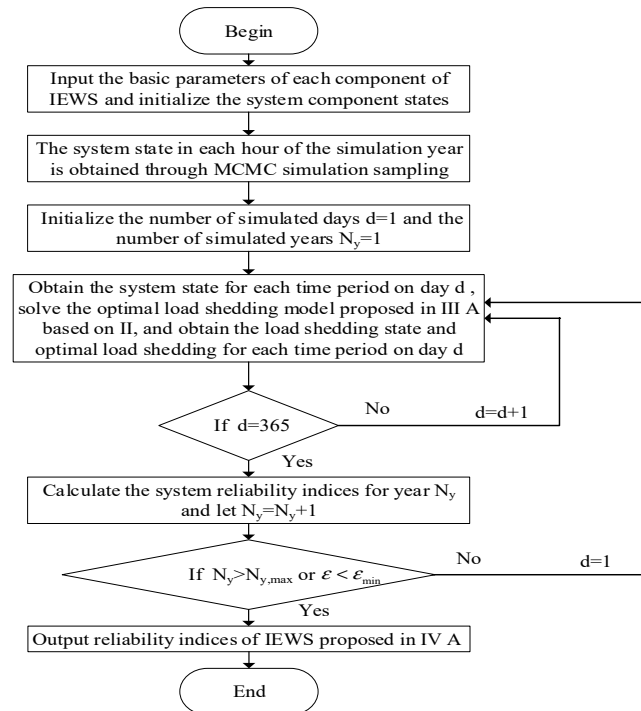
$$CENS = c_p EENS + c_w EWNS + c_l EWL \quad (36)$$

where  $c_p$ ,  $c_w$ , and  $c_l$  denote the penalty cost per unit of electric load shedding, water load shedding, and pipeline leakage, respectively; and EENS, EWNS, and EWL de-

note the expected values for electric load shedding, water load shedding, and pipeline leakage, respectively.

#### 4.2. Reliability Evaluation Methods

In this section, the reliability evaluation approach is developed, based on the MCMC simulation methodology, by considering the WS equipment's temporal state transfer process [22]. The specific calculation steps are shown in Figure 7.



**Figure 7.** Flow chart of the IEWS reliability evaluation algorithm.

## 5. Case Study

Two test IEWSs, a 6-node power system with a 21-node WS and an IEEE-30-node power system with a 36-node WS, are used to verify the proposed model, the detailed data of the two systems can be found in Ref. [7]. It should be noted that the integrated electricity–water system in this article belongs to the same operator, and the data privacy issues between different operators are not applicable. First, test system A (6-node power system and 21-node WS) illustrates the advantages of the multi-state reliability evaluation approach proposed in this paper. Then, test system B (IEEE 30-node power system and 36-node WS) is used to illustrate the adaptability of the model and reliability evaluation proposed in this paper to different IEWSs. The penalty cost per unit for electric load shedding, water load shedding, and pipeline leakage are 71.43 USD/MWh, 1.14 USD/kg, and 1.14 USD/kg, respectively.

### 5.1. IEWS Test A

- (1) Optimization results of the IEWS considering the equipment failures of the WS.

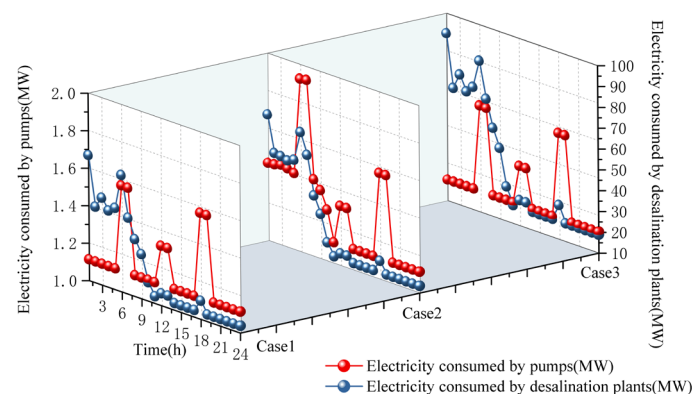
In this section, three cases are set up to illustrate the impact of equipment failures in the WS on the operation of the IEWS.

Case1: Water supply pipes, RO desalination plants, tanks, and valves operate normally.

Case2: RO desalination plants, tanks, and valves operate normally; the water supply pipe I21 has a 10 mm hole leakage which fails from 1:00 a.m. to 11:00 a.m. and is repaired at 11:00 a.m.

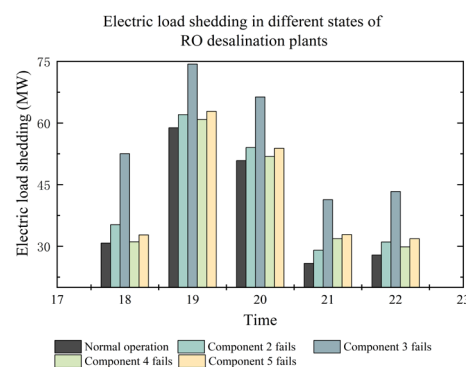
Case3: The water supply pipes, tanks, and valves operate normally; the RO desalination plant's component 3 constituents are inefficient.

Figure 8 represents the three cases of energy consumption for the pumps and desalination plants. From the figure, it can be seen that the power consumption of the water pumps and desalination plant in Case2 are higher than in Case1 from 1:00 a.m. to 11:00 a.m., which means when there is a leakage in the pipeline. The power consumption of the water pumps increases significantly to satisfy the users' demand for water as much as possible. The power consumption of the pumps in Case3 is the same as that in Case1. However, the power consumption of the desalination plant in Case3 is significantly higher than that in Case1 and Case2, because the desalination plant is in an inefficient operating condition and consumes more power to produce the same quality of fresh water.



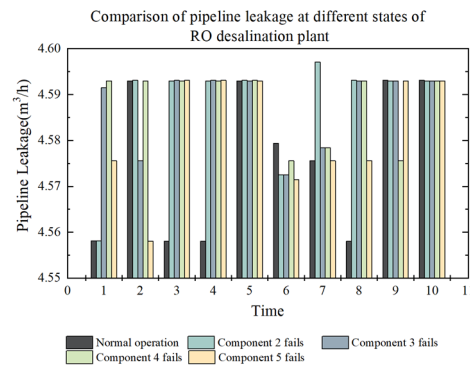
**Figure 8.** Energy consumption of water pumps and RO desalination plants.

Figure 9 shows the electrical load shedding at every hour in different inefficient operation states of the RO desalination plant based on Case2 and assuming the failure of the generator set G1. During normal operation of the RO desalination plant, the total electrical load shedding is 194.16 MW, assuming the thermal generator G1 fails. From Figure 9, the electrical load shedding in different inefficient operating states of the RO desalination plant can all be seen to be higher than the shedding during normal operation.



**Figure 9.** The electrical load shedding in different inefficient operating states of the RO desalination plant.

Figure 10 represents the pipe leakage at each moment in different states of the RO desalination plant. Since the pipeline leakage was repaired from 11:00 a.m. to 24:00 p.m., the figure shows the pipeline leakage from 1:00 a.m. to 10:00 a.m. The figure shows that when the desalination plant is operating inefficiently, the total pipeline leakage increases in all cases.



**Figure 10.** Pipeline leakage in different inefficient operating states of RO desalination plant.

## (2) System reliability analysis

For the IEWS test A, the following four models are used to illustrate the advantages of the multi-state model equipment in the WS proposed in this paper.

Model 1: The two-state model of water supply pipes and the RO desalination plant is adopted.

Model 2: The two-state model of water supply pipes and the multi-state model of the RO desalination plant are adopted.

Model 3: The multi-state model of water supply pipes and the two-state model of the RO desalination plant are adopted.

Model 4: The proposed multi-state model of water supply pipes and the RO desalination plant has been adopted.

The reliability evaluation indices of the IEWS for the four models are shown in Table 1.

**Table 1.** Reliability evaluation indices of case A.

| Reliability Indices         | Model 1            | Model 2            | Model3             | Model 4            |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|
| LOELP                       | 0.0036             | 0.0100             | 0.0055             | 0.01311            |
| EENS (MWh/year)             | 833.32             | 2626.08            | 2455.03            | 3005.88            |
| LOWLP                       | 0.0047             | 0.0041             | 0.0049             | 0.0031             |
| EWNS (m <sup>3</sup> /year) | 44,833.17          | 20,737.16          | 46,540.79          | 14,338.015         |
| PWL                         | 0.00024            | 0.00028            | 0.001178           | 0.00130            |
| EWL (m <sup>3</sup> /year)  | 505.59             | 677.25             | 1279.377           | 1379.589           |
| C <sub>ENS</sub> (USD/year) | $1.11 \times 10^5$ | $2.11 \times 10^5$ | $2.30 \times 10^5$ | $2.33 \times 10^5$ |

Compared with Model 1, the LOELP, EENS, EWL, PWL, and C<sub>ENS</sub> of Model 2 increase, and the LOWLP and EWNS of Model 2 decrease. This is because the RO desalination plant's multi-state operating features are considered in Model 2. More electric energy will be consumed when the RO desalination plant is in an inefficient operation state, increasing the operation pressure of the power system and decreasing its reliability. For the WS, when the desalination plant operates inefficiently, it can still supply fresh water to the users of the WS to reduce water load shedding, so the reliability of the WS will still be improved, although the probability and expected value of the leakage will increase.

Compared with Model 1, the LOELP, EENS, LOWLP, EWNS, PWL, EWL, and C<sub>ENS</sub> of Model 3 increase. This is due to the fact that the multi-state operation model of the pipeline is considered in Model 3. When the water supply pipe is in the leakage state, the power consumption of the pumps will increase to meet the water demand of the users, and the overall power consumption of the WS increases, as well as the operational pressure of the power system, thus leading to a decrease in the reliability of the power system. At the same time, compared with the fault state, pipeline leakage is not easy to detect, and maintenance

costs are relatively high, resulting in the inability to repair the pipeline on time, resulting in more serious leakage and thus reducing the level of reliability of the water system.

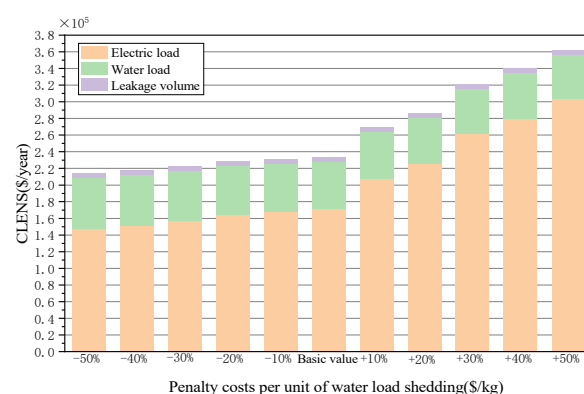
Compared with Model 1, the reliability level of the power system in Model 4 is significantly reduced, the reliability of the water system is improved, and the probability and expected value of pipeline leakage are increased. It can be seen that the two-state model of water supply pipes and the RO desalination plant overestimate the reliability of the power system in the IEWS. At the same time, ignoring pipeline leakage, the operation of the WS cannot be accurately simulated.

Compared with Model 2, the LOELP, EENS, PWL, and EWL of Model 4 increase. This is because the multi-state operation model of the pipeline is considered in Model 4. In order to meet user needs when there is a pipeline leakage, the water pump must be in a high-power operation state, increasing the operating pressure of the power system and reducing its reliability. This shows that only considering the multi-state of the RO desalination plant and ignoring the existence of pipeline leakage will still overestimate the reliability of the power system. Compared with Model 3, the LEOLP, EENS, PWL, and EWL of Model 4 all increase. This is because the RO desalination plant's multi-state operation features are considered in Model 4. This shows that when the RO desalination plant operates inefficiently, the energy consumption of producing the same quality of water increases, putting operating pressure on the power system. In addition, due to the continuous water supply, pipeline leakage will also increase. This shows that only considering the two-state of the water supply pipeline and ignoring the inefficient operation of the RO desalination plant will overestimate the reliability of the power system. Therefore, it is necessary to consider the multi-state model of the RO desalination plant and the water supply pipeline at the same time.

In summary, the multi-state reliability model of the water supply pipes and RO desalination plant proposed in this paper can fully consider the existence of leakage in the water supply pipes and the electric load shedding caused by the inefficient operation of the seawater desalination plant. Thus, they can avoid overestimating the reliability level of the power system during the joint operation of the IEWS.

### (3) Impact of water load shedding penalty cost

In order to analyze the influence of  $c_w$  on  $C_{ENS}$ , the magnitude of  $c_w$  is changed on the basis of Method 4, the change trend of  $C_{ENS}$  is observed, and the results are shown in Figure 11.



**Figure 11.** Combined loss of energy shortage in different water load shedding penalty costs.

As can be seen from Figure 11, when  $c_w$  is lower than the base value, the system chooses to shed the water load as much as possible in order to reduce electric load shedding in the case of equipment failure, such as power lines, thus reducing the total load shedding

penalty cost. As  $c_w$  gradually decreases, the level of electric load reliability increases slightly and  $C_{ENS}$  keeps decreasing. When  $c_w$  is higher than the base value, the system chooses to reduce the electric load as much as possible to ensure the reliability of the water supply when the desalination plant is operating inefficiently. As  $c_w$  increases, the load reliability level decreases and  $C_{ENS}$  increases.

### 5.2. IEWS Test B

In order to further verify the adaptability of the model proposed in this paper as well as the reliability evaluation methods, the four models set up in test A for the system reliability analysis are used to assess the reliability of system B. The simulation results are shown in Table 2.

**Table 2.** Reliability evaluation indices of case B.

| Reliability Indices         | Model 1            | Model 2            | Model3             | Model 4            |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|
| LOELP                       | 0.00282            | 0.00574            | 0.00292            | 0.00655            |
| EENS (MWh/year)             | 1589.35            | 4017.03            | 3991.23            | 5941.64            |
| LOWLP                       | 0.0076             | 0.0072             | 0.0081             | 0.0067             |
| EWNS (m <sup>3</sup> /year) | 148,200.01         | 68,864.99          | 145,967.66         | 37,373.33          |
| PWL                         | 0.00072            | 0.00084            | 0.00333            | 0.00391            |
| EWL (m <sup>3</sup> /year)  | 1892.53            | 2758.65            | 5167.88            | 5936.14            |
| $C_{ENS}$ (USD/year)        | $2.85 \times 10^5$ | $3.69 \times 10^5$ | $4.58 \times 10^5$ | $4.59 \times 10^5$ |

Compared with Method 1, the LOELP, EENS, and  $C_{ENS}$  of Method 4 increase, while the LOWLP and EWNS of Method 4 decrease. In summary, the simulation conclusions in this section are consistent with IEWS test A. Using the multi-state reliability model of the equipment in the WS proposed in this paper for the reliability evaluation of the IEWS can avoid overestimating the reliability level of the power system, and the IEWS reliability evaluation method proposed in this paper is also applicable to large systems.

## 6. Conclusions

Access to new energy sources such as wind power and photovoltaic power poses severe challenges to the reliability of the power systems, and the reliability problems of the WS are critical due to the failure of the water pipes and other components. Since component failures can propagate in integrated systems, the reliability problem of the IEWS becomes more prominent when the two systems are strongly coupled. This paper investigates the reliability of the IEWS based on the multi-state models of the equipment in the WS. First, a multi-state Markov model for the water pipes is established, based on the pipe failure mechanism, to analyze the impact of pipe leakage on the operation state of the IEWS. Second, based on the failure mechanism of the RO desalination plant, a multi-state Markov model is established to analyze the influence of inefficient operation on the operation state of the IEWS. Finally, the IEWS reliability evaluation indices and method are proposed to quantify the reliability level and economic loss of the power and water supply in the IEWS, which can provide key decision-making support for system planning and construction. The case study results show that ignoring the leakage of the water pipes and the inefficient operation of the RO desalination plant overestimates the level of power supply reliability. The multi-state reliability evaluation model of the equipment in the WS can simulate the actual operation of the IEWS, and this can more accurately evaluate the reliability level of the power and water supply in the IEWS. In addition, this paper also verifies that with an increase in the penalty cost per unit of water load, the reliability level of the water supply increases, but the reliability level of the power supply decreases gradually. Therefore, it is

essential to set a reasonable penalty cost for the reduction in water supply to balance the reliability of the power system and the economic operation of the IEWS.

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